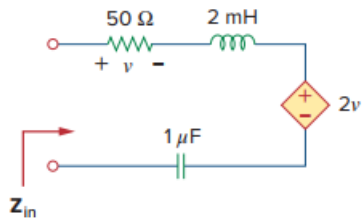


Problem

For the circuit in Fig. 9.69, find the input impedance Z_{in} at 10 krad/s.



Step-by-step solution

Step 1 of 4

Refer to Figure 9.69 in the textbook.

Calculate the impedance of inductor.

$$Z_L = j\omega L$$

Substitute 10×10^3 for ω , and 2×10^{-3} for L .

$$Z_L = j(10 \times 10^3)(2 \times 10^{-3})$$

$$Z_L = j20 \Omega$$

Calculate the impedance of capacitor.

$$Z_C = \frac{1}{j\omega C}$$

Substitute 10×10^3 for ω , and 10^{-6} for C .

$$Z_C = \frac{1}{j(10 \times 10^3)(1 \times 10^{-6})}$$

$$Z_C = -j100 \Omega$$

Step 2 of 4

Connect a test source $1 \angle 0^\circ \text{A}$ to the given circuit as shown below.

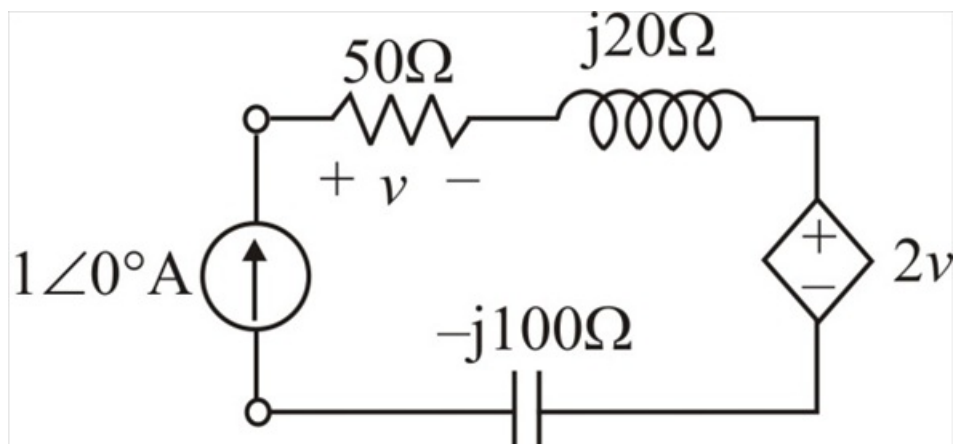


Figure 1

Step 3 of 4

From the above circuit, we can write,

$$v = 50(1\angle 0^\circ)$$

$$v = 50V$$

Assume that the voltage across $1\angle 0^\circ A$ current source is V_{1A} .

Apply Kirchhoff's voltage law.

$$V_{1A} - 1\angle 0^\circ(50 + j20 - j100) - 2v = 0$$

$$V_{1A} = 1\angle 0^\circ(50 + j20 - j100) + 2v$$

Substitute 50 for v .

$$V_{1A} = 1\angle 0^\circ(50 - j80) + 2(50)$$

$$V_{1A} = 50 - j80 + 100$$

$$V_{1A} = (150 - j80) V$$

Step 4 of 4

Calculate the input impedance Z_{in} .

$$Z_{in} = \frac{V_{1A}}{1\angle 0^\circ}$$

Substitute $(150 - j80)$ for V_{1A} .

$$Z_{in} = \frac{150 - j80}{1\angle 0^\circ}$$

Therefore, the value of Z_{in} is $(150 - j80) \Omega$.